

Algebra I Linear Functions Cheatsheet

Module 5

Slope

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

positive slope: increasing, negative slope: decreasing

zero slope: horizontal line, undefined slope: vertical line

Slope-Intercept Form

$$y = mx + b$$

$$m = \text{slope}, \quad b = y\text{-intercept}$$

Graph by plotting $(0, b)$, then using rise over run.

Point-Slope Form

$$y - y_1 = m(x - x_1)$$

Use when you know one point and the slope.

Standard Form

$$Ax + By = C$$

Intercepts:

$$x\text{-intercept: set } y = 0, \quad y\text{-intercept: set } x = 0$$

Parallel and Perpendicular Lines

Parallel lines have the same slope:

$$m_1 = m_2$$

Perpendicular lines have slopes that are opposite reciprocals:

$$m_1 m_2 = -1$$

Function Notation

$$f(x) = mx + b$$

$f(3)$ means substitute $x = 3$

Writing Linear Equations

- From slope and intercept: use $y = mx + b$.
- From slope and point: use $y - y_1 = m(x - x_1)$.
- From two points: find m , then use point-slope form.
- From a table: check whether first differences are constant.

Quick Checks

- Slope is change in y over change in x .
- Vertical lines are $x = a$ and are not functions.
- Horizontal lines are $y = b$ and have slope 0.
- The y -intercept occurs when $x = 0$.